L4b: Single Asset Geometric Brownian Motion Models

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Objectives for Today

By the end of this lecture, you can:

- Describe and simulate GBM: Explain drift and random fluctuations, interpret the price distribution, and simulate prices using the exact solution.
- Estimate and interpret GBM parameters: Estimate mean growth and volatility from historical prices, recover the drift, and estimate mean growth using log-price regression.
- Assess GBM and calculate trade probabilities: Compare GBM with observed growth-rate behavior and calculate the probability of exceeding a target scaled NPV at a scheduled sale time.

Lectures and Examples

Lecture: Single-Asset Geometric Brownian Motion

Example: Estimating GBM Parameters
Estimate mean growth rate and volatility from historical prices, recover the arithmetic drift, and compare simulated price paths with the data.

Example: GBM Terminal-Target Probabilities
Compute the probability that a long stock position exceeds a target scaled NPV at a scheduled sale time.

Company Profile: Jane Street

Jane Street is a privately held quantitative trading firm. Like Citadel Securities in L4a, it uses mathematical models and software in trading.

- Trading business: Jane Street trades mainly with its own capital and provides liquidity to institutional clients. Citadel Securities also executes orders for retail brokers and institutional investors.
- Products: Jane Street is known for market making in exchange-traded funds (ETFs), which let investors trade baskets of assets. Both firms operate across multiple asset classes.
- Technology: Jane Street uses OCaml, a functional programming language, in its trading systems.

A trade's outcome depends on price changes before the sale. We will model this uncertainty with GBM and calculate the probability of exceeding a target scaled NPV.

From Lattices to Continuous Prices

In L4a, lattice models used discrete time steps and a finite set of price movements. What changes as we reduce the time step?

- Keep the holding period fixed and divide it into more steps.
- Shrink the up and down price movements and adjust their probabilities so the mean and variance of the holding-period log return remain approximately unchanged.
- For suitably scaled, independent binomial steps, the accumulated log return $\ln(S_T/S_0)$ approaches a normal distribution.

Since $S_T = S_0 \exp[\ln(S_T/S_0)]$, the terminal price approaches a lognormal distribution. The corresponding continuous-time price model is geometric Brownian motion.

We introduce this model through its stochastic differential equation, then use its solution to calculate a trade's target probability.

Single Asset Geometric Brownian Motion (GBM)

Paul Samuelson (1965) used geometric Brownian motion to model stock prices. The model became widely known through Black–Scholes–Merton (BSM) option pricing. Random fluctuations scale with the current price, and prices remain positive.

Let $S(t) > 0$ be the share price (USD/share), with t and the holding period T measured in years. The constant parameters are drift $\mu \in \mathbb{R}$ (year^{-1}) and volatility $\sigma > 0$ (year$^{-1/2}$). With Wiener increment $dW(t)$, GBM is given by:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu\, dt}_{\text{drift}} + \underbrace{\sigma\, dW(t)}_{\text{noise}}, \qquad 0 \leq t \leq T.$$

The drift sets the expected proportional change per unit time. The Wiener increment supplies the random shock. Let's define this noise process.

The Wiener Process

A Wiener process $\{W(t),\, 0 \leq t \leq T\}$ is a continuous real-valued stochastic process with three properties:

- The process starts at zero: $W(0) = 0$ with probability one.
- Increments over disjoint intervals $W(t_1) - W(t_0), \dots, W(t_k) - W(t_{k-1})$ are independent for any $0 \leq t_0 < t_1 < \cdots < t_k \leq T$.
- Each increment is normal with mean zero and variance equal to the elapsed time, $W(t) - W(s) \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t \leq T$.

The third property lets us write any increment as a scaled standard normal:

$$W(t) - W(s) = \sqrt{t - s}\, Z, \qquad Z \sim \mathcal{N}(0, 1).$$

The Analytical Solution

Take $t_0 = 0$ and let S_0 be today's price. Solving the SDE with Itô calculus gives the share price at time T:

$$\boxed{S_T = S_0 \exp\Big[\underbrace{(\mu - \tfrac{\sigma^2}{2})T}_{\text{mean log growth}} + \underbrace{\sigma\sqrt{T}\,Z}_{\text{shock}}\Big], \qquad Z \sim \mathcal{N}(0,1).}$$

Thus the log price ratio $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and variance $\sigma^2 T$, and the price S_T is lognormal and strictly positive.

Itô calculus gives the half-variance correction $-\sigma^2/2$. This makes the mean growth rate differ from the drift μ. Next, we compute the mean and variance of future prices.

Derivation: Deriving the GBM Solution

Mean and Variance of the GBM Price

From the lognormal moments, the expectation and variance of S_T are:

$$\underbrace{\mathbb{E}(S_T)}_{\text{mean price}} = S_0 e^{\mu T}, \qquad \underbrace{\mathrm{Var}(S_T)}_{\text{price variance}} = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right).$$

Properties:

- The mean averages over possible GBM outcomes. It grows exponentially when $\mu > 0$ and declines when $\mu < 0$, but individual price paths can rise or fall around this expectation.
- The variance describes the spread of prices: $S_0^2 e^{2\mu T}$ is the squared expected price, while $e^{\sigma^2 T} - 1$ captures the effect of volatility over the holding period.

For fixed initial price, drift, and holding period, increasing volatility widens the price distribution without changing the expected price.

Discrete-Time GBM Model

To simulate a price path, use a grid $t_j = j\Delta t$, $j = 0, 1, \dots, N$, with time step Δt (years) and horizon $T = N\Delta t$. Applying the GBM solution between successive times gives the one-step transition:

$$\boxed{S_{t_j} = S_{t_{j-1}} \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_j\right].}$$

Here, $j = 1, \dots, N$, and $Z_1, \dots, Z_N$ are independent standard normal random variables. Their independence follows from the independent Wiener increments over successive intervals.

For constant μ and σ, this transition gives exact GBM prices at the grid points. We use it to generate price paths with Monte Carlo simulation.

Monte Carlo Simulation of GBM

Given S_0, μ, σ, Δt, N steps, and M paths, store the simulated prices in $\mathbf{S} \in \mathbb{R}^{(N+1)\times M}$, one path per column, rows indexed by $j = 0, \dots, N$.

For each sample path (each column of $\mathbf{S}$):

1. Set the price in row $j = 0$ to S_0.
2. For $j = 1, \dots, N$: draw $Z_j \sim \mathcal{N}(0, 1)$ and apply the one-step transition:

$$S_{t_j} \leftarrow S_{t_{j-1}} \exp\Big[(\mu - \tfrac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}\, Z_j\Big].$$

Each step of each path uses a fresh independent draw. The result is M price trajectories of N steps each.

Let's estimate the drift and volatility needed for simulation from historical share prices.

Estimation of GBM Parameters

To connect GBM to data, divide the one-step transition by $S_{t_{j-1}}$, take logarithms, and divide by Δt. This gives the continuously compounded growth rate g_j (year^{-1}) used in L3a:

$$\boxed{g_j \equiv \frac{1}{\Delta t}\ln\frac{S_{t_j}}{S_{t_{j-1}}} = \underbrace{\left(\mu - \frac{\sigma^2}{2}\right)}_{\text{mean growth } \mu_g} + \underbrace{\frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate std}} Z_j.}$$

Because the Z_j are independent standard normals, growth rates over equally spaced, non-overlapping steps are independent and normal with mean $\mu_g \equiv \mu - \sigma^2/2$ and variance $\sigma^2/\Delta t$.

- Drift versus growth: μ is the SDE drift; the mean growth rate μ_g is estimated from a growth-rate series. They differ by the half-variance correction.
- Volatility from dispersion: the growth-rate standard deviation is $\sigma/\sqrt{\Delta t}$, while the one-step log-return standard deviation is $\sigma\sqrt{\Delta t}$. We use either to estimate σ.

Volatility

From $N+1$ prices we form N growth rates $g_1, \ldots, g_N$. Their sample mean g' and sample standard deviation σ_g (our L3a risk measure) are:

$$g' = \frac{1}{N}\sum_{j=1}^{N} g_j, \qquad \underbrace{\sigma_g}_{\text{growth-rate std}} = \sqrt{\frac{1}{N-1}\sum_{j=1}^{N}\left(g_j - g'\right)^2}.$$

Estimating the mean leaves $N-1$ degrees of freedom. To recover GBM volatility from σ_g, use $\mathrm{Var}(g_j) = \sigma^2/\Delta t$: σ_g estimates $\sigma/\sqrt{\Delta t}$, giving the volatility estimate:

$$\boxed{\hat{\sigma} = \sigma_g\sqrt{\Delta t} = \sigma_{\text{ann}}.}$$

This matches L3a's annualized volatility σ_{ann}, with units year$^{-1/2}$.
For daily data $\Delta t = 1/252$: divide the daily growth-rate standard deviation by $\sqrt{252}$, or equivalently multiply the daily log-return standard deviation by $\sqrt{252}$.

Mean Growth and Drift

With volatility estimated, we need mean growth to recover the drift. Since $\mu = \mu_g + \sigma^2/2$, an estimate $\hat{\mu}_g$ gives:

$$\underbrace{\hat{\mu}}_{\text{SDE drift}} = \underbrace{\hat{\mu}_g}_{\text{mean growth}} + \underbrace{\tfrac{1}{2}\hat{\sigma}^2}_{\text{correction}}.$$

Two common ways to estimate μ_g:

- Method 1, average growth rate: $\hat{\mu}_g = g'$, the sample mean and maximum likelihood estimate under GBM.
- Method 2, linear regression: fit log price as a line in time. The slope estimates μ_g.

Let's estimate mean growth by regressing log price on time.

Linear Regression: The Model

Suppose we observe prices $S_{t_0}, \ldots, S_{t_N}$ at times $t_j = j\Delta t$. Taking the logarithm of the GBM solution gives:

$$\ln S_{t_j} = \underbrace{\ln S_0}_{\text{intercept}} + \underbrace{\mu_g t_j}_{\text{trend}} + \underbrace{\sigma W(t_j)}_{\text{error}}, \qquad j = 0, \ldots, N.$$

Since the Wiener process has mean zero, the expected log price is a line with intercept $\ln S_0$ and slope μ_g. Least squares fits this line by minimizing the sum of squared differences from the observed log prices.

The error $\sigma W(t_j)$ contains all shocks up to t_j, so errors at different times share shocks and are correlated. Their variance is $\sigma^2 t_j$. These properties affect uncertainty in the fitted intercept and slope.

Linear Regression: The Normal Equations

To fit the intercept and slope together, stack the $N+1$ log prices as $\hat{\mathbf{X}}\theta + \epsilon = \mathbf{y}$, where $\theta = (\ln S_0, \mu_g)^\top$ contains the free intercept and slope:

$$\hat{\mathbf{X}} = \begin{bmatrix} 1 & t_0 \\ 1 & t_1 \\ \vdots & \vdots \\ 1 & t_N \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} \ln S_{t_0} \\ \ln S_{t_1} \\ \vdots \\ \ln S_{t_N} \end{bmatrix}.$$

For full column rank (at least two distinct observation times), the least-squares solution is given by:

$$\boxed{\hat{\theta} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}.}$$

The first component estimates the intercept. The second gives the mean-growth estimate $\hat{\mu}_g$, which we combine with volatility to recover $\hat{\mu} = \hat{\mu}_g + \hat{\sigma}^2/2$.

Regression Intervals: Standard Errors

For comparison with Brownian errors, assume independent normal errors with mean zero and common variance σ_ϵ^2.

For $n = N + 1$ observations and $p = 2$ fitted parameters, the residuals and error variance estimate are:

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{X}}\hat{\theta}, \qquad \hat{\sigma}_\epsilon^2 = \frac{\|\mathbf{r}\|_2^2}{n - p}.$$

With $n - p$ degrees of freedom after fitting, the standard error is given by:

$$\boxed{\mathrm{SE}(\hat{\theta}_j) = \sqrt{\hat{\sigma}_\epsilon^2 \left[(\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}\right]_{jj}}, \qquad j = 1, 2.}$$

Standard error measures uncertainty in the fitted intercept or mean growth rate, in the parameter's units. Smaller values indicate greater precision.

Regression Intervals: Confidence Intervals

We use the standard error to set an interval around each estimate. Under the independent normal-error assumptions, a confidence interval with level $1-\alpha$ for θ_j is given by:

$$\boxed{\hat{\theta}_j \pm t_{1-\alpha/2,\nu}\,\mathrm{SE}(\hat{\theta}_j), \qquad \nu = n - p.}$$

Here, $t_{1-\alpha/2,\nu}$ is the Student t quantile with ν degrees of freedom. For a 95% confidence interval, set $\alpha = 0.05$, giving the multiplier $t_{0.975,\nu}$. Here, $j = 1$ selects the intercept and $j = 2$ mean growth.

The advanced drift-uncertainty example develops parameter uncertainty under GBM and examines its effect on target probabilities. Let's apply our estimates and ordinary regression intervals to historical share-price data.

Example: Estimating GBM Parameters

Stylized Facts: Heavy-Tailed Growth Rates

Does fitted GBM reproduce L3a's patterns: heavy tails, little linear autocorrelation, and volatility clustering? Recall the model's growth-rate law:

$$g_j = \mu_g + \frac{\sigma}{\sqrt{\Delta t}} Z_j, \qquad Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

With constant parameters and independent normal draws, one-step growth rates are normal.

Heavy tails: observed growth rates have heavier tails than a normal distribution. The model therefore assigns too little probability to extreme movements.

The lognormal price has a long right tail, but growth rates depend on the log price ratio. A lognormal price does not give heavy-tailed growth rates.

Stylized Facts: Little Linear Autocorrelation

From the growth-rate law, $g_j = \mu_g + (\sigma/\sqrt{\Delta t})Z_j$ with independent shocks. The autocovariance between growth rates $k \geq 1$ observation intervals apart is given by:

$$\mathrm{Cov}(g_j, g_{j+k}) = \frac{\sigma^2}{\Delta t}\,\mathrm{Cov}(Z_j, Z_{j+k}) = 0,$$

because Z_j and Z_{j+k} are independent. With $\mathrm{Var}(g_j) = \sigma^2/\Delta t$, the autocorrelation function is:

$$\boxed{\rho_g(k) = \frac{\mathrm{Cov}(g_j, g_{j+k})}{\mathrm{Var}(g_j)} = 0, \qquad k \geq 1.}$$

This agrees with the weak linear dependence observed in signed growth rates. The model assumes independence, which is stronger than zero autocorrelation. Let's examine what independence implies for growth-rate magnitudes.

Stylized Facts: Volatility Clustering

Volatility clustering concerns the magnitudes of growth rates. Large absolute movements tend to follow large movements, while small movements tend to follow small ones. This produces periods of high and low volatility in observed data.

Under GBM, σ is constant and shocks are independent. A large absolute growth rate therefore does not make another large movement more likely at the next step. The model cannot reproduce volatility clustering.

Across these three patterns, GBM agrees with weak linear autocorrelation but misses heavy tails and volatility clustering. Let's use its analytical price distribution to evaluate a scheduled trade.

GBM Trade Rule

Consider a scheduled trade: buy $n_0 > 0$ shares at S_0 at time 0 and sell at S_T at time $T = N\Delta t$. Assume no dividends or trading costs.

Let g_y be the constant, continuously compounded benchmark growth rate (year^{-1}) corresponding to yield y. The NPV is given by:

$$\text{NPV}(g_y, T) = \underbrace{-\,n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted sale}}.$$

Dividing by the initial investment gives the scaled NPV, our discounted fractional return:

$$\rho_T = \frac{\text{NPV}(g_y, T)}{n_0 S_0} = \left(\frac{S_T}{S_0}\right) e^{-g_y T} - 1.$$

We now use GBM to describe the uncertain sale price S_T.

Substitute the GBM Solution

Using mean growth rate $\mu_g = \mu - \sigma^2/2$, substitute the GBM price into the scaled NPV:

$$\boxed{\rho_T = \exp\Big[(\mu_g - g_y)T + \sigma\sqrt{T}\,Z\Big] - 1, \qquad Z \sim \mathcal{N}(0,1).}$$

Discounting subtracts the benchmark growth rate g_y from the mean growth rate μ_g, while leaving the random shock unchanged. The quantity $1 + \rho_T$ is lognormal, so:

$$\ln(1 + \rho_T) \sim \mathcal{N}\big((\mu_g - g_y)T, \sigma^2 T\big)\,.$$

The scaled NPV is a lognormal variable shifted down by one, with $\rho_T > -1$.

The benchmark affects the return through $g_y T$. When $|g_y|T \ll 1$, discounting is small and $\rho_T \approx S_T/S_0 - 1$. We use the exact discounted expression to calculate the target probability.

Terminal Target Probability

Choose a target scaled NPV $\rho_\star > -1$. Adding one and taking the logarithm preserves the strict target condition:

$$\rho_T > \rho_\star \quad \Longleftrightarrow \quad \ln(1+\rho_T) > \ln(1+\rho_\star).$$

Substituting the GBM expression gives:

$$(\mu_g - g_y)T + \sigma\sqrt{T}\,Z > \ln(1+\rho_\star).$$

Subtracting $(\mu_g - g_y)T$ and dividing by $\sigma\sqrt{T} > 0$ gives a threshold for the standard normal random variable Z:

$$Z > z_\star, \qquad \boxed{z_\star = \frac{\ln(1+\rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}.}$$

Thus $Z > z_\star$ means the trade exceeds its target at the scheduled sale. Selling when a boundary is first reached requires a separate calculation.

The Closed-Form Target Probability

The standard normal cumulative distribution function $\Phi(z_\star)$ gives the probability that $Z \leq z_\star$. Subtracting from one gives the probability that $Z > z_\star$:

$$\boxed{\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi(z_\star) = 1 - \Phi\left(\frac{\ln(1+\rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}\right).}$$

- For a chosen target, we evaluate $z_\star$ from estimated mean growth and volatility, the holding period, and benchmark growth rate.
- The formula applies for $T > 0$, $\sigma > 0$, and $\rho_\star > -1$.
- A target of -1 or below is exceeded with probability one because the scaled NPV always exceeds -1 under GBM.

Example: GBM Terminal-Target Probabilities

Optional Advanced Material

Advanced: From the Binomial Lattice to GBM
Calibrate the lattice to GBM and track convergence of its terminal distribution and target probability.

Advanced: First-Passage Rules Under GBM
Compare daily and continuous monitoring for single- and two-barrier exit rules.

Advanced: Drift Uncertainty
Examine how mean-growth precision depends on the observation period and affects a target probability.

Advanced: Monte Carlo Target Probabilities
Compare exact GBM and Euler–Maruyama simulation, quantify sampling uncertainty, and introduce variance reduction techniques.

Optional; none of these notebooks is a prerequisite for L5a.

Summary

Today we developed a continuous-time share-price model, estimated its parameters, and used its price distribution to evaluate a stock trade.

Key takeaways

- Continuous-time price modeling: We used the exact GBM solution to describe lognormal prices and simulate paths. It captures zero growth-rate autocorrelation but misses heavy tails and volatility clustering.
- Mean growth, drift, and volatility: We estimated mean growth from averages and log-price regression, and volatility from growth-rate dispersion. The half-variance correction recovers drift: $\mu = \mu_g + \sigma^2/2$.
- Terminal target probabilities: We used the GBM price distribution to calculate the probability of exceeding a target scaled NPV at the scheduled sale time, accounting for the holding period and benchmark growth rate.

Next, we extend GBM to correlated assets and introduce portfolio covariance.